

Credit Default & Financial Risk Analysis

Portfolio Performance & Default Segmentation Report • Active Portfolio: 300 Overdue Contracts

Executive Summary

This analysis identifies client profiles with higher default risk and measures the total financial exposure of the portfolio. The primary objective is to provide actionable data for the Risk Management department, enabling the optimization of credit recovery strategies and the mitigation of future losses.

TOTAL OVERDUE AMOUNT	TOTAL OVERDUE CONTRACTS	AVERAGE OVERDUE VALUE
€2,239,480	300	€7,464.93

1. Introduction

The analysis focuses on identifying delinquency patterns and risk segmentation by loan type and income bracket. With a portfolio comprising 300 contracts in arrears, this report details where the institution's highest financial vulnerabilities are concentrated, establishing quantitative benchmarks for recovery operations.

2. Portfolio Overview & Key Metrics

The foundational metrics reflecting the current financial health and delinquency distribution of the analyzed portfolio:

Indicator / Metric	Result / Value	Validation & Notes
Total Default Amount	€2,239,480	Sum of all active overdue contracts
Average Overdue Value	€7,464.93	Valid: (€2,239,480 ÷ 300 ≈ €7,464.93)
Maximum Overdue Value	€14,759.00	Highest single contract exposure
Minimum Overdue Value	€56.00	Lowest single contract exposure

Indicator / Metric	Result / Value	Validation & Notes
Average Days in Arrears	192.92 days	Coherent distribution across 1 to 365 days

Loan Type Distribution

Loan Type	No. of Loans
Personal Loan	88
Mortgage	67
Auto Loan	54
Credit Card	48
Student Loan	25
Business Loan	18
Total	300

Detailed Portfolio Insights

Insight 1: Financial Exposure The portfolio presents a total exposure of €2,239,480 in defaulted credit, subject to recovery actions.
Insight 2: Delinquency Severity Average overdue value of €7,464.93 demonstrates high individual risk concentration.
Insight 3: Portfolio Composition Personal Loans dominate volume (88), followed by Mortgages (67) and Auto Loans (54).

3. Risk & Demographic Analysis

Correlation between credit categories, monthly income, age groups, and rigorous financial risk classifications.

Average Overdue by Loan Type

- 1. **Auto Loan:** €8,137
- 2. **Personal Loan:** €7,881
- 3. **Business Loan:** €7,636

Average Days in Arrears

- 1. **Business Loan:** 200.33 days
- 2. **Personal Loan:** 199.00 days
- 3. **Credit Card:** 198.00 days

Income Bracket Breakdown

Income Bracket	Clients	Avg. Overdue (€)
High Income	131	€7,380.73
Medium Income	113	€7,514.90
Low Income	56	€7,561.07

*Note: Italy records the highest country average (€10,221.92), followed by France, with Portugal registering the lowest.

Age Group Delinquency Analysis

Age Group	Total Customers	Average Days Overdue
18–30 years	50	191.50 days
31–45 years	79	193.72 days
46–60 years	85	189.04 days

Age Group	Total Customers	Average Days Overdue
Over 60 years	86	196.86 days

4. Customer Risk Scoring Model

A multi-factor risk classification model was established integrating **Days in Arrears**, **Overdue Amount**, and **Monthly Income**.

HIGH RISK

≥ 180 days overdue
AND ≥ €7,000 debt
AND Income < €3,000

MEDIUM RISK

90–179 days overdue
OR Debt €4K–€6.9K
OR Income €3K–€4K

LOW RISK

Remaining clients meeting neither high nor medium thresholds.

Top 14 High-Exposure Customers (Sample Extract)

Customer Name	Country	Income (€)	Loan Type	Amount Overdue (€)
Catarina Oliveira	Portugal	2,400	Personal Loan	12,225
Carlos Carvalho	Portugal	1,650	Personal Loan	12,537
Miguel Mendes	Portugal	2,000	Personal Loan	12,698
Bruno Pinto	Portugal	2,300	Personal Loan	11,008
Miguel Costa	Portugal	1,600	Student Loan	14,142
Sofia Sousa	França	2,700	Credit Card	10,071
Tiago Lopes	Portugal	2,250	Personal Loan	9,839
Luis Gomes	Portugal	2,850	Mortgage	8,413
Inês Rodrigues	Portugal	2,700	Mortgage	10,518
Maria Teixeira	Portugal	2,350	Personal Loan	11,865
Inês Ferreira	Portugal	1,400	Personal Loan	9,816
Inês Martins	Portugal	2,850	Mortgage	7,834
Catarina Rocha	Portugal	1,500	Credit Card	10,807
André Ferreira	Brasil	2,400	Auto Loan	9,535

Annex: SQL Queries

The queries implemented in Google BigQuery to extract and categorize portfolio metrics:

Query 1: Total Default Amount

```
SELECT SUM(amount_overdue) AS total_overdue_amount
FROM `luis-melo-468018.credit_default_analysis.loan_default_project`;
```

Query 2: Average Overdue Value

```
SELECT AVG(amount_overdue) AS valor_medio_atraso
FROM `luis-melo-468018.credit_default_analysis.loan_default_project`;
```

Query 3: Customer Risk Classification (Top 20)

```
SELECT CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
       c.country, c.monthly_income, l.loan_type, l.amount_overdue, l.days_overdue,
       CASE
         WHEN l.days_overdue >= 180 AND l.amount_overdue >= 7000 AND c.monthly_income < 3000 THEN
           'High Risk'
         WHEN l.days_overdue BETWEEN 90 AND 179 OR l.amount_overdue BETWEEN 4000 AND 6999 OR
           c.monthly_income BETWEEN 3000 AND 4000 THEN 'Medium Risk'
         ELSE 'Low Risk'
       END AS risk_level
FROM `luis-melo-468018.credit_default_analysis.customers_project` AS c
INNER JOIN `luis-melo-468018.credit_default_analysis.loan_default_project` AS l ON c.customer_id =
l.customer_id
ORDER BY CASE WHEN risk_level = 'High Risk' THEN 1 WHEN risk_level = 'Medium Risk' THEN 2 ELSE 3 END,
         l.days_overdue DESC, l.amount_overdue DESC
LIMIT 20;
```